

QAISUMAH, Saudi Arabia

WMO: 403730

Lat: 28.335N

Lon: 46.125E

Elev: 1174

StdP: 14.08

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 38.2	(c) 41.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 24.4	(k) 56.8	(l) 21.6	(m) 57.3	(n) 3.8	(o) 270	(p) 0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 26.4	(c) 116.4	(d) N/A	(e) 114.6	(f) N/A	(g) 112.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.3	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.5	(b) 19.8	(c) 17.2	(e) 32.6	(f) 119.4	(g) 3.8	(h) 2.6	(i) 29.9	(j) 121.3	(k) 27.7	(l) 122.8	(m) 25.5	(n) 124.3	(o) 22.8	(p) 126.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.9	54.7	59.3	68.0	78.7	90.0	96.0	98.7	99.0	93.0	83.5	67.2	57.2	
	DBStd	17.04	5.82	6.40	6.77	6.46	5.12	3.55	3.49	3.46	4.07	5.41	7.01	6.11	
	HDD50	36	20	6	0	0	0	0	0	0	0	0	0	10	
	HDD65	854	322	179	43	2	0	0	0	0	0	0	57	251	
	CDD50	10570	166	266	558	860	1240	1379	1509	1518	1290	1037	515	232	
	CDD65	5912	3	18	136	412	775	929	1044	1053	840	572	122	8	
	CDH74	96152	47	190	1430	5019	12415	16528	19078	19063	13856	7355	1081	90	
CDH80	67588	6	42	561	2678	8381	12239	14617	14607	9727	4334	386	10		
Wind	WSAvg	7.6	7.0	7.8	8.4	8.8	8.3	8.4	8.1	7.1	6.2	7.0	6.8	6.8	
Precipitation	PrecAvg	3.8	0.7	0.6	0.3	0.6	0.1	0.0	0.0	0.0	0.0	0.2	0.7	0.7	
	PrecMax	9.1	2.1	2.0	2.0	2.1	0.7	0.2	0.0	0.0	0.0	1.6	3.5	3.8	
	PrecMin	1.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	2.1	0.6	0.6	0.6	0.5	0.2	0.0	0.0	0.0	0.0	0.4	1.1	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	85.8	97.3	104.4	112.5	116.4	118.6	118.5	114.5	105.9	93.3	80.9	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	73.6	79.2	91.3	100.4	109.1	113.8	116.4	116.6	111.5	102.5	88.2	76.6	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	68.8	75.3	86.0	96.8	106.2	111.5	114.5	114.7	109.2	100.2	84.5	71.9	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	65.8	71.6	81.5	93.2	104.0	109.5	112.0	112.7	106.7	97.1	80.6	68.1	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	MCDDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	Clear-Sky Solar Irradiance	taub	0.464	0.516	0.580	0.668	0.690	0.626	0.610	0.589	0.551	0.544	0.523	0.472	
		taud	1.833	1.717	1.589	1.473	1.447	1.530	1.579	1.620	1.678	1.692	1.732	1.824	
Ebn at Noon		235	235	229	215	211	224	227	231	234	223	214	225		
Edh at Noon		56	69	83	96	99	91	86	82	75	70	62	55		
All-Sky Solar Radiation	RadAvg	1193	1476	1802	1977	2233	2506	2422	2278	2036	1642	1217	1087		
	RadStd	79	81	75	82	107	82	63	48	57	72	97	89		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.97	N/A	N/A	+1.21	N/A	N/A	N/A	+373	+306
(47) Regional (4 neighbors)		+0.97	N/A	N/A	+1.21	N/A	N/A	N/A	+373	+306

Nomenclature: See separate page